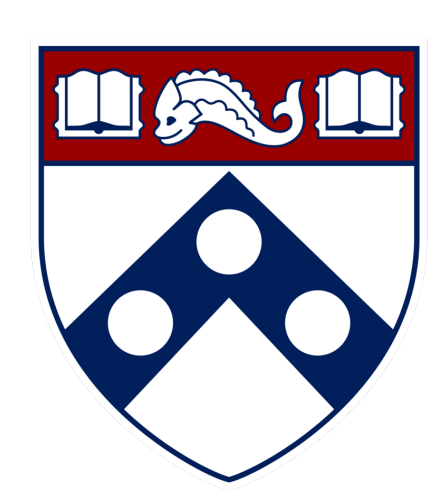




Transformers Learn the Optimal DDPM Denoiser for Multi-Token GMMs

Hongkang Li
UPenn

Hancheng Min
SJTU

René Vidal
UPenn



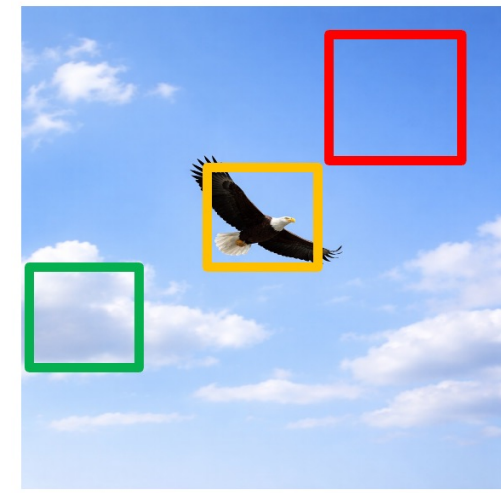
Overview

- Diffusion models** generate samples by learning either
 - a **denoiser** $f(\Psi; X, t)$: parameter Ψ , input X , diffusion time step t ;
 - the **score function**: $s(X, t) = \nabla_X \log p(X)$.
- Gap between theory and practice**: existing work studies
 - Landscape: **no proof of convergence** to optimal denoiser.
 - Training dynamics: **guarantees only for simplified architectures**.
- Contributions**: for **transformer-based diffusion models** trained on **multi-token Gaussian mixture data** with **population loss**, we show that gradient descent converges to **the Bayes-optimal denoiser**.

Data Model and Problem Formulation

Motivations:

Image tokens correspond to a small number of patterns.



- Tokens of the sky**: $\mathcal{N}(\mu_1, \rho^2 I)$.
- Tokens of the eagle**: $\mathcal{N}(\mu_2, \rho^2 I)$.
- Tokens of the cloud**: $\mathcal{N}(\mu_3, \rho^2 I)$.

Data Model: Multi-token Gaussian mixture (MTGM)

- $X = (x_1, \dots, x_P) \in \mathbb{R}^{d \times P} \sim \mathcal{D}(\tilde{\pi}, K, \{\mu_i\}_{i=1}^M, \rho)$. Mixture of M components.
- Select $K \leq M$ components**: $Z \sim \text{Unif}(z \in \{0, 1\}^M : 0 < \|z\|_0 = K)$.
- Normalize proportions**: $\pi(Z)$ with $\pi_i(Z) = [Z]_i \tilde{\pi}_i / Z^\top \tilde{\pi}$.
- Sample P tokens**: $x_p \mid Z \sim \text{MoG}(\pi(Z), \{\mu_i\}_{i=1}^M, \rho^2 I)$.

Forward Process: (DDPM) $X^t = \sqrt{\alpha_t} X^0 + \sqrt{1 - \alpha_t} E^t$, $E^t \sim \mathcal{N}(0, I)$, $t \in [T]$.

Scheduled noise

Gaussian noise

Denoising Network: $f(\Psi; X, t) = v_t(X - X \text{softmax}(X^\top W X / d))$, $\Psi = \{W, \{v_t\}_{t=1}^T\}$.

- DDPM population loss** $L_D(\Psi) = \sum_{t=1}^T \mathbb{E}_{X^0, E^t} \left[\|f(\Psi; X^t, t) - E^t\|_F^2 / (2dPT) \right]$.

- Score matching** $L_S(\Psi) = \sum_{t=1}^T \mathbb{E}_{X^0, E^t} \left[\|g(\Psi, X^t, t) - s(X^t, t)\|_F^2 / (2dPT) \right]$.

Gradient updates: $\Psi^{(\tau+1)} = \Psi^{(\tau)} - \eta \nabla_\Psi L(\Psi^{(\tau)})$.

Main Theoretical Results

Notations

- Time-averaged SNR: $SNR = \mathbb{E}_t[\bar{\alpha}_t / (1 - \bar{\alpha}_t)]$.
- Minimal pattern ratio: $\nu(K) = \min_{u \in [M]} \mathbb{P}(\text{a token is drawn from pattern } u) \leq \frac{1}{M}$.
- Bayes denoising risk: $R_{\text{Bayes}} = \mathbb{E}_{X^0, E^t, t} \left[\|\mathbb{E}[E^t | X^t] - E^t\|_F^2 / (2dP) \right]$.

Theorem 1: Convergence of DDPM Training.

For any $\epsilon \in (0, \min_{u \in [M]} \tilde{\pi}_u^{\Theta(1)})$, if the step size $\eta \leq O(1)$,

- the token dimension $d \geq \epsilon^{-1} \log(\nu(K)^{-1})$
- the number of iterations $I = \Omega((\epsilon^{-1} + \nu(K)^{-3}) \eta^{-1} \nu(K)^{-1} \cdot SNR^{-3})$,
- the number of tokens per data $P \geq \Omega(\nu(K)^{-1} (\rho^2 + 1) \epsilon^{-1} \log d)$,
- the number of diffusion steps $T \geq \Omega(\log d)$,

then with high probability, the learned model $\Psi^{(I)}$ satisfies $L(\Psi^{(I)}) \leq R_{\text{Bayes}} + O(\epsilon)$.

Corollary 1: Special Cases of the Required Number of Iterations I .

- Small I : $K = 1$, $I = \Omega((\epsilon^{-1} + M^3) \eta^{-1} M \cdot SNR^{-3})$;
- Large I : $K = M$, $I = \Omega((\epsilon^{-1} + (\min_{u \in [M]} \tilde{\pi}_u)^{-3}) \eta^{-1} (\min_{u \in [M]} \tilde{\pi}_u)^{-1} \cdot SNR^{-3})$.

Theorem 2: Score Matching.

Let $\Psi^{(I)}$ be defined as in Theorem 1. For some $\epsilon \in (0, \min_{u \in [M]} \tilde{\pi}_u^{\Theta(1)})$, we can construct the following score network with parameters $\Psi^{(I)}$

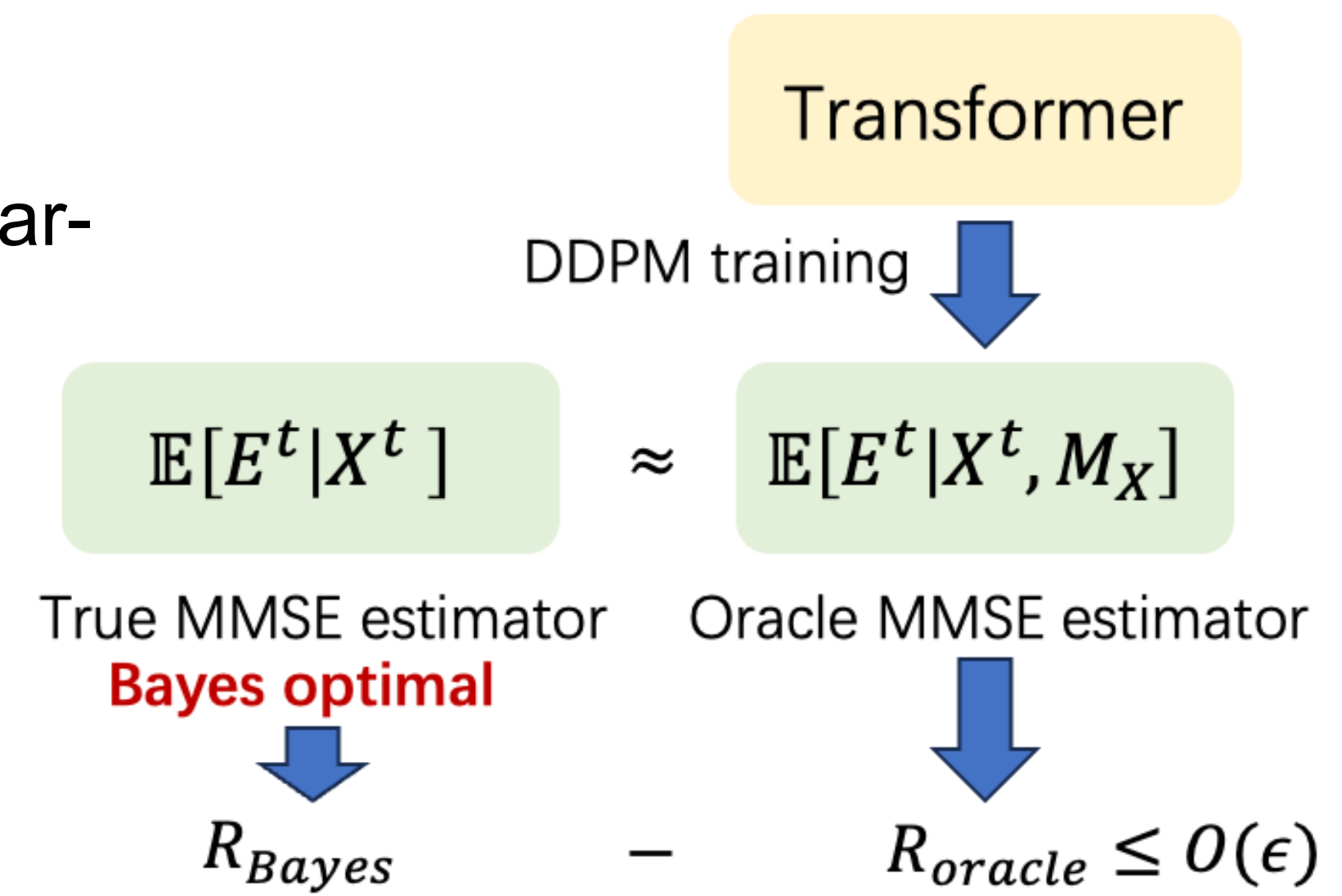
$$g(\Psi^{(I)}, X^t, t) = -f(\Psi^{(I)}; X^t, t) / \sqrt{1 - \bar{\alpha}_t},$$

such that $L_S(\Psi^{(I)}) \leq L_D(\Psi^{(I)}) \cdot (SNR + 1) \leq \epsilon \cdot (SNR + 1)$.

Denoising Mechanism of Transformers

Key Insights

The trained Transformer achieves near-optimal denoising by **approximating the oracle MMSE estimator**.



Notations

- M_X : the patterns of all tokens of X .
- Oracle denoising risk: $R_{\text{oracle}} = \frac{1}{2dP} \mathbb{E}_{X^0, E^t, t} \left[\|\mathbb{E}[E^t | X^t, M_X] - E^t\|_F^2 \right]$.

Proposition 1: Transformer Learns Oracle MMSE Estimator

$$\begin{aligned} \text{Trained Model} &= v_t^{(I)} \left(X^t - X^t \text{softmax} \left(\frac{X^{t\top} W^{(I)} X^t}{d} \right) \right) \\ &\updownarrow O(\epsilon) \\ \text{Oracle MMSE} &= \frac{\sqrt{1 - \bar{\alpha}_t}}{1 - \bar{\alpha}_t + \rho^2 \bar{\alpha}_t} X^t - \sqrt{\bar{\alpha}_t} M_X \end{aligned}$$

How can $X^t \text{softmax}(\frac{X^{t\top} W^{(I)} X^t}{d})$ approximate $\sqrt{\bar{\alpha}_t} M_X$?

Proposition 2: Mean Denoising Mechanism

Attention weights are concentrated and uniformly distributed among tokens with the same pattern as the query, leading to a minimum-variance unbiased estimator (MVUE) of the mean pattern of each token for approximation.

